

Multi-Layer Perceptron (MLP) Implementation Report

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Abstract. This report presents the implementation and analysis of a Multi-Layer Perceptron (MLP) model. The model is trained using selected hyperparameters and evaluated through loss curves and prediction results. The findings highlight how tuning parameters affects model performance.

1 Introduction

MLP is a fundamental neural network used for solving classification and regression problems. This project focuses on understanding its learning process.

2 Background

MLP consists of layers of neurons with activation functions such as ReLU and Sigmoid. Learning occurs using backpropagation and gradient descent.

3 Implementation

The model is implemented using forward propagation and backpropagation to update weights iteratively.

4 Experiments

4.1 Dataset

The dataset contains input features and corresponding outputs used for training and testing.

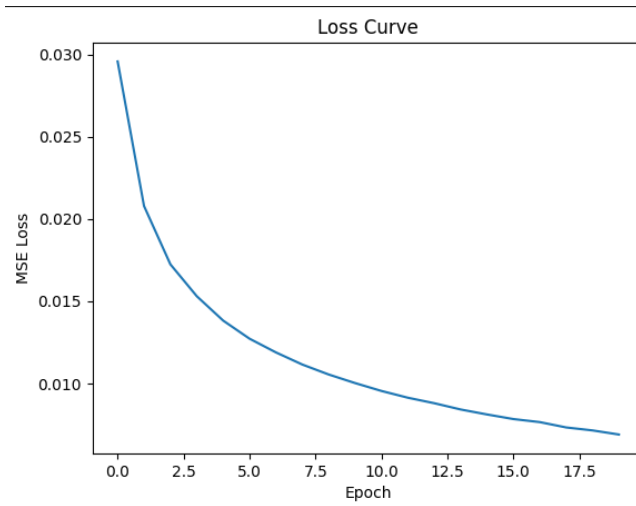
Table 1. Hyperparameters Used

Parameter	Value
Learning Rate	0.1
Epochs	20
Batch Size	32

Table 2. Training Loss per Epoch

Epoch Loss	
1	0.0296
2	0.0208
3	0.0172
4	0.0153
5	0.0138
6	0.0127
7	0.0119
8	0.0111
9	0.0105
10	0.0100
11	0.0095
12	0.0091
13	0.0088
14	0.0084
15	0.0081
16	0.0078
17	0.0076
18	0.0073
19	0.0071
20	0.0069

4.2 Loss Curve

**Fig. 1.** Loss curve over training epochs

5 Discussion

The results show that decreasing loss improves accuracy. Proper tuning of hyperparameters is essential for better performance.

6 Conclusion

The MLP model successfully learns patterns from data. This project demonstrates the importance of training techniques and parameter tuning.

References

1. Goodfellow, I., Bengio, Y., Courville, A.: Deep Learning. MIT Press (2016)
2. Haykin, S.: Neural Networks and Learning Systems. Pearson (2009)